

The impact of cow's milk-mediated mTORC1-signaling in the initiation and progression of prostate cancer

Prostate cancer (PCa) is dependent on androgen receptor signaling and aberrations of the PI3K-Akt-mTORC1 pathway mediating excessive and sustained growth signaling. The nutrient-sensitive kinase mTORC1 is upregulated in nearly 100% of advanced human PCas. Oncogenic mTORC1 signaling activates key subsets of mRNAs that cooperate in distinct steps of PCa initiation and progression. Epidemiological evidence points to increased dairy protein consumption as a major dietary risk factor for the development of PCa. mTORC1 is a master regulator of protein synthesis, lipid synthesis and autophagy pathways that couple nutrient sensing to cell growth and cancer. This review provides evidence that PCa initiation and progression are promoted by cow's milk, but not human milk, stimulation of mTORC1 signaling. Mammalian milk is presented as an endocrine signaling system, which activates mTORC1, promotes cell growth and proliferation and suppresses autophagy. Naturally, milk-mediated mTORC1 signaling is restricted only to the postnatal growth phase of mammals. However, persistent consumption of cow's milk proteins in humans provide highly insulinotropic branched-chain amino acids (BCAAs) provided by milk's fast hydrolysable whey proteins, which elevate postprandial plasma insulin levels, and increase hepatic IGF-1 plasma concentrations by casein-derived amino acids. BCAAs, insulin and IGF-1 are pivotal activating signals of mTORC1. Increased cow's milk protein-mediated mTORC1 signaling along with constant exposure to commercial cow's milk estrogens derived from pregnant cows may explain the observed association between high dairy consumption and increased risk of PCa in Westernized societies. As well-balanced mTORC1-signaling plays an important role in appropriate prostate morphogenesis and differentiation, exaggerated mTORC1-signaling by high cow's milk consumption predominantly during critical growth phases of prostate development and differentiation may exert long-term adverse effects on prostate health. Attenuation of mTORC1 signaling by contemporary Paleolithic diets and restriction of dairy protein intake, especially during mTORC1-dependent phases of prostate development and differentiation, may offer protection from the most common dairy-promoted cancer in men of Western societies.